



A method to extract the Quaternion Ship Domain parameters from AIS data

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ABSTRACT

The ship domain concept is of great interest for ship traffic modelling, risk assessment and intelligent collision avoidance. The paper proposes and applies a method to define AIS data-based empirical polygonal ship domains, based on traffic density matrices that are derived around each reference ship from an AIS dataset. A modified Quaternion Ship Domain allowing for different shapes for each quadrant is proposed, which results in a better fitting to the empirical domain. The parameters of Quaternion Ship Domains that best fit the empirical polygonal domains are determined for cargo ships and tankers of different lengths. Violations of the Quaternion Ship Domain are then used as an indicator of collision risk that is graphically represented in the study area, providing important information to support maritime traffic monitoring and control tasks.

1. Introduction

Although maritime transportation contributes significantly to the world's economic growth, it also involves risks related to accidental events such as fires, groundings and collisions, among others (Guedes Soares and Teixeira, 2001). Ship collisions accounted for 26.2% of the total number of marine casualties with ships between 2011 and 2018, according to the European Maritime Safety Agency (EMSA) (EMSA, 2019), and can potentially result in disastrous consequences for the environment and the economic activities dependent on the sea, such as fisheries and tourism. Most of the accidents are caused by human errors at different levels (Antão and Guedes Soares, 2019; Shi et al., 2021; Fan et al., 2020; Coraddu et al., 2020), but other aspects related to the environmental (Antão and Guedes Soares, 2010) and the navigational conditions at some local sea areas also contribute to the maritime casualties (e.g. Silveira et al., 2013; Rong et al., 2021). In fact, it is well known that maritime accidents result from the combined effect of several technological/technical, human, or organizational causes under a socio-technical system, as discussed by Salihoglu and Bal Beşikçi (2021) and Lee and Chung (2018).

The need to improve the safety of navigation has led to the implementation of coastal and port Vessel Traffic Services (VTS), that rely on several systems to provide monitoring, control and information services. Among these systems, the Automatic Identification System (AIS) has become an important source of information for studying maritime traffic and associated risks. This system allows the automatic exchange of

information between stations (ships and coastal), using VHF radio waves. From 31 December 2004, all vessels over 300 GT engaged in international voyages, all cargo vessels over 500 GT and all passenger vessels are required to be equipped with AIS. A ship's position, course, heading, speed, dimensions, type, draught, destination and other important data are broadcasted at regular intervals by AIS transceivers and can be stored for later analysis, making AIS data a valuable resource to researchers.

AIS data requires decoding and pre-processing before it can be used in research. Decoding is a straightforward process, although deviations from the standard exist and should be anticipated when developing decoding software. Several types of errors have been found in AIS messages, some due to poor human interaction during installation and operation of the AIS equipment, others due to poor integration with other navigational equipment such as compasses, others yet due to noise contamination (Harati-Mokhtari et al., 2007) that may prevent the use of AIS for collision avoidance (Stitt, 2004). Pre-processing AIS messages to address these and other errors is an important step to obtain meaningful results from data.

Tu et al. (2018) have provided a state-of-the-art overview on the use of AIS in maritime transportation research, covering the historical and real-time AIS data sources, as well as the methodologies that use AIS data for route estimation, collision prediction and anomaly detection.

A great amount of research work exists on the topic of route or trajectory identification and its applications to AIS data, resulting in a variety of methods for route extraction, prediction and anomaly

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detection (Etienne et al., 2010; Pallotta et al., 2013; Rong et al., 2019, 2020a; Silveira et al., 2019; Vespe et al., 2012); ; ; ; . In particular, safe path identification is an important step in the development of certified safe routes that have the potential to become one of the key enablers for future autonomous vessel operations and may facilitate route exchange mechanisms, thus playing a role in future methods to reduce collision risk (Zhang et al., 2021; Xu et al., 2019).

An important use of AIS data has been on the development of collision risk assessment models and to identify possible traffic conflicts such as collision candidates (Pedersen, 1995; Silveira et al., 2013) or the so-called “near collisions” (Zhang et al. 2015, 2016). These non-accidental events are observed more frequently in AIS datasets than actual collisions, thus providing more information to study the influence of the maritime traffic characteristics on the traffic conflicts (Rong et al., 2021) and to derive geographical distributions of the collision risk in several sea areas (Yoo, 2018; Rong et al., 2020b).

The concept of ship domain was introduced by Fujii and Tanaka (1971), and since then it has been widely used as a near-collision criterion in several maritime traffic safety studies. A well-known definition of ship domain formulated by Goodwin (1975), is described as “the surrounding effective waters which the navigator of a ship wants to keep clear of other ships or fixed objects”.

The ship domain concept has been frequently combined with collision risk indices such as the DCPA (Distance at Closest Point of Approach) and the TCPA (Time to Closest Point of Approach) or more sophisticated indices such as the Minimum Distance To Collision (MDTC) proposed in Montewka et al. (2010), to detect the near collisions and to assess the collision risk of ship-ship encounters (e.g. Rong et al., 2020b; Zhang et al., 2015, 2016; Mou et al., 2010).

When studying ship collision risk, one cannot ignore the International Regulations for Preventing Collisions at Sea, commonly referred to as the COLREGs. The COLREGs may be considered subjective (Kuwata et al., 2014), in the sense that the same scenarios may be interpreted differently and may produce different collision risk evaluations and, consequently, different avoidance actions by navigators following the same set of rules. Thus, assessing ship collision risk can be approached as a decision problem, where experts take the place of decision-makers, who must assign a risk class to a given ship encounter scenario, as proposed by Silveira et al. (2021).

Even the applicability or the moment when the rules begin to apply have been the motive for interesting discussions (Zhao, 2008). Although Rule 7 of the COLREGs addresses the risk of collision, no definition is included and no guidance on how to assess the risk is provided, other than the statement that “*such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change*”. Different interpretations of the rules, along with possibly other factors, may result in different perceptions of safety, which in turn are likely to have an impact on the ship domain.

Goerlandt et al. (2015) identified three underlying mechanisms in interpreting risk: the imminence of an accident occurrence; a deviation from what is considered a reference level in similar situations; and the possibility that another agent in the situation interprets it differently, which can also be described as the possibility of a situation entailing ambiguity. Over the next Chapters of this paper, AIS data is used to obtain empirical ship domains and the parameters of an analytical Quaternion Ship Domain model that best fit them. A modification of the Quaternion Ship Domain is proposed, that includes different shape parameters for each quadrant, therefore allowing for a better fit to generic empirical ship domain shapes. The obtained analytical ship domains are used to assess collision risk, exploring links between the concept of ship domain and the previously mentioned mechanisms used to interpret risk, particularly the deviation from a reference safety level.

2. Ship domain

An extensively used concept when studying maritime traffic safety is

the ship domain, an area around the vessel which the navigator would like to keep free (or, by some definitions, actually keeps free) of other vessels, for safety reasons (Fujii and Tanaka, 1971; Goodwin, 1975; Coldwell, 1983). This concept has been used in marine traffic engineering and is of great interest for ship traffic modelling and risk assessment, as well as for designing bridges or other obstacles to marine traffic (Hansen et al., 2013).

Wang (2013) stated that ship domains are powerful and fundamental approaches for dealing with ship navigation risk assessment, collision avoidance, marine traffic simulations and optimal trajectory planning. This author divides the innovation of ship domains into three methodologies: statistical, individual and intelligent models, where statistical models focus on macro identification for common ship domains, individual models depend on specific ship parameters and dimensions, and intelligent models, which can be considered an approximation of statistical models, use learning algorithms to identify ship domains from collected data. Wang et al. (2009) proposed a unified analytical framework for three existing typical ship domain shapes: circular, elliptical and polygonal. These researchers present mathematical frameworks for each ship domain types and perform a validation of the analytical domain models and a comparative study for different ship domains, using computer simulations on different encounter situations, concluding that the analytical ship domain models are effective.

Some shipping companies and ship masters require their officers to keep the distance to other vessels over a minimum value, often designated as a minimum Closest Point of Approach (CPA) and stated in the navigation standing orders. This requirement can be thought of as a circular-shaped ship domain. By defining a circular ship domain, a vessel's master or navigator is implicitly stating that the minimum required distance from other ships, for safety reasons, is the same in every direction. However, in practice, this may not be the case. In fact, in their extensive review of ship domains, Szlapczynski and Szlapczynska (2017) stated that the introduction of the ship safety domain to maritime navigation comes from the observation that the safe distance is not the same in all directions. These authors present three classical definitions of ship domain (Fujii and Tanaka, 1971; Goodwin, 1975; Coldwell, 1983) and analyse four safety criteria resulting from different interpretations of those definitions, which have been used by different researchers:

- own ship's (OS) domain should not be violated by a target ship (TS)
- a TS's domain should not be violated by the OS
- neither of the ship domains should be violated
- ship domains should not overlap

Szlapczynski and Szlapczynska (2017) stated that ship domains can be determined empirically, based on experts' knowledge or based on safety analysis, presenting relevant examples for each group and adding that the three groups are not mutually exclusive, since combinations of two or three approaches are possible. These authors point out some limitations of each group: empirical ship domains are dependent on the traffic density and on the percentage of head-on, crossing and overtaking encounters, since this can vary according to the characteristics of the water area (e.g. open seas, narrow waterways, TSS); knowledge-based ship domains are dependent on the choice of surveyed navigators or experts and the choice of the survey questions and scenarios; analysis-based domains are also subjective, as the results rely on assumptions (which may be disputable) and factors which depend on researchers' preferences.

Fiskin et al. (2020) introduced a knowledge-based model structure to create an asymmetrical polygonal ship domain for both restricted waters and open waters. An application based on a Mandani type fuzzy inference system is implemented to determine the size and shape of the domain. The proposed model is compared with other ship domains to evaluate its performance and is found to be more conservative than others.

Among the analysis-based ship domain models, the Quaternion Ship

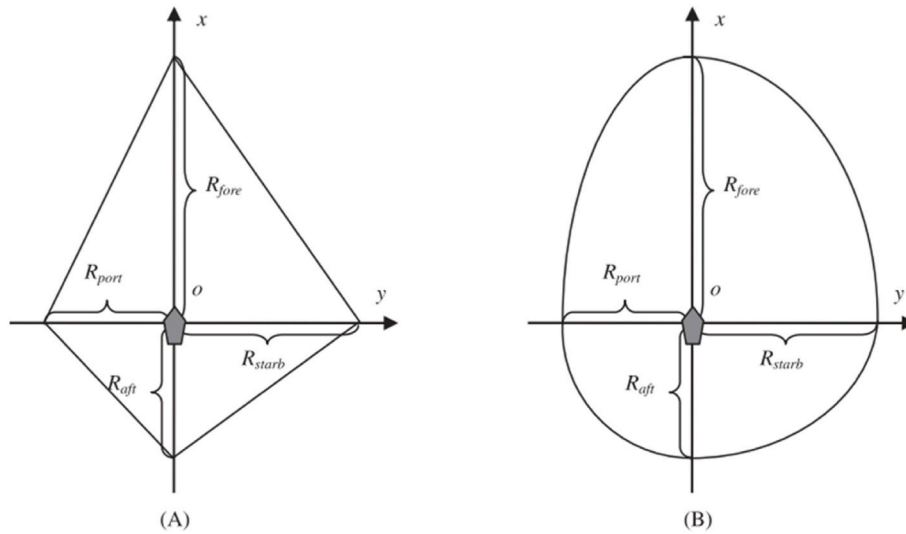


Fig. 1. Quaternion Ship Domain, $k = 1$ (A), $k = 2$ (B) (Wang, 2010).

Domain (Wang, 2010) is of particular interest and is presented as dependable and flexible for navigators to use to make decisions. The QSD's size is determined by the quaternion Q including four radii (R_{fore} , R_{aft} , R_{starb} and R_{port}) and its shape is modelled by parameter k . Equation (1) defines the QSD that is graphically represented in Fig. 1, for $k = 1$ (A) and $k = 2$ (B).

$$f_k(x, y; Q) = \left(\frac{2x}{(1 + \text{sgn } x)R_{fore} - (1 - \text{sgn } x)R_{aft}} \right)^k + \left(\frac{2y}{(1 + \text{sgn } y)R_{starb} - (1 - \text{sgn } y)R_{port}} \right)^k \quad (1)$$

Wang (2010) presented estimation formulae for parameters of the blocking area (Kijima and Furukawa, 2003) as references to estimate longitudinal and lateral radii in Q :

$$\begin{cases} R_{fore} = \left(1 + 1.34 \sqrt{k_{AD}^2 + \left(\frac{k_{DT}}{2} \right)^2} \right) L \\ R_{aft} = \left(1 + 0.67 \sqrt{k_{AD}^2 + \left(\frac{k_{DT}}{2} \right)^2} \right) L \\ R_{starb} = (0.2 + k_{DT})L \\ R_{port} = (0.2 + 0.75k_{DT})L \end{cases} \quad (2)$$

where L is the own ship length, k_{AD} and k_{DT} represent gains of the advance A_D and of the tactical diameter D_T respectively, and can be calculated as follows, with V_{own} as the own ship speed in knots:

$$\begin{cases} k_{AD} = A_D/L = 10^{0.3591 \lg V_{own} + 0.0952} \\ k_{DT} = D_T/L = 10^{0.5441 \lg V_{own} - 0.0795} \end{cases} \quad (3)$$

According to Wang (2010), ship domains are affected by factors such as manoeuvring capability, dimensions, courses and speeds of encountered ships, and COLREG rules. These factors can be sufficiently taken into account by the parameters of the QSD. This researcher proposes the Fuzzy Quaternion Ship Domain (FQSD) model, an extension of the QSD with a series of boundaries to indicate different levels of navigational safety, as an appropriate model to assess collision risk. A Spatial Collision Risk (SCR) function, composed of Longitudinal Collision Risk and Lateral Collision Risk membership functions, is defined. It is concluded that the proposed QSD, FQSD and SCR can be used for collision risk assessment and collision avoidance.

The impact of ship size and speed on the ship domain has been

studied by Pietrzykowski and Wielgosz (2021). The boundaries of empirical ship domains were obtained by analysing the outcome of simulations performed by nautical officers in a training simulator. Six encounter situations representing the most typical events were selected for the simulations. The ship domains' boundaries were determined using a method of partial domains. These authors conclude that the ship size has a considerable impact on the ship domain shape and size, whereas the impact of ship speed is found to be smaller.

As previously mentioned, AIS data has been used in numerous areas of research, including for studying ship domains. Hansen et al. (2013) used observations from southern Danish waters over four years to establish an empirical ship domain. The ship domain shape is analysed by visualizing distances, measured from each reference ship to all other ships, in intensity plots. A threshold of 3500 m is set, meaning that only distances up to this value are included in the analysis. All distances are normalized for the length of the reference ship. The intensity plots generated reflect the amount of time during which target ships are observed at a given position relative to the reference ship. These researchers found the shape of the ship domain to be elliptic, and determined the length of the ellipse in a narrow TSS and when crossing a ferry route. They also analysed the distance in the breadth direction in two distinct geographical areas with different characteristics. The researchers concluded that the empirical elliptic domain for that area has a length of 8 ship lengths and a breadth of 3.2 ship lengths, which is compatible with the ship domain proposed by Fujii and Tanaka (1971).

Wang and Chin (2016) calibrated free-form ship domains of an asymmetrical polygonal shape, which size is assumed to be dynamically enlarged with increasing ship speeds, using the data of vessel movements in Singapore Strait and Singapore Port. These researchers state that although statistical methods have been initially used to determine ship domains, most ship domains have not been well calibrated. Three shortcomings of statistical methods for obtaining ship domains, which have been identified by Pietrzykowski and Uriasz (2009), are presented: the necessity for sufficient amount of data; difficulties in separating the factors affecting the ship domain shape and size; an unclear description of the perceived clear area around the ship.

Wang and Chin (2016) classified navigational waterways into three categories: open sea waterways; confined waterways; and narrow fairways and channels, stating that traffic densities in these three categories are significantly different. These researchers propose a domain in which each vertex of a polygon is a function of the ship's Length Over All (LOA) and speed. A speed function governs how the domain is expanded. Calibration is done by estimating the function parameters which optimize the distance between the model behaviour and the data concerning

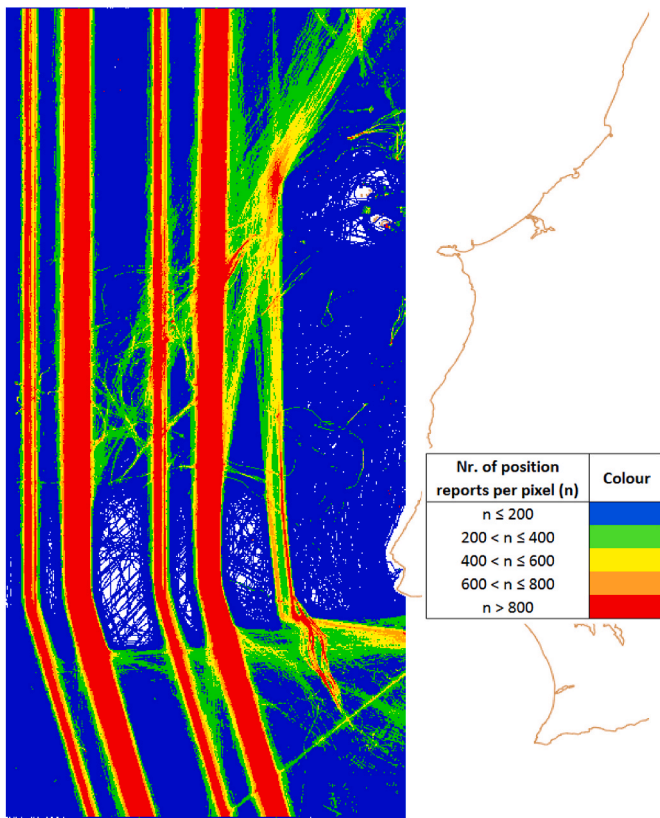


Fig. 2. AIS vessel position reports density map.

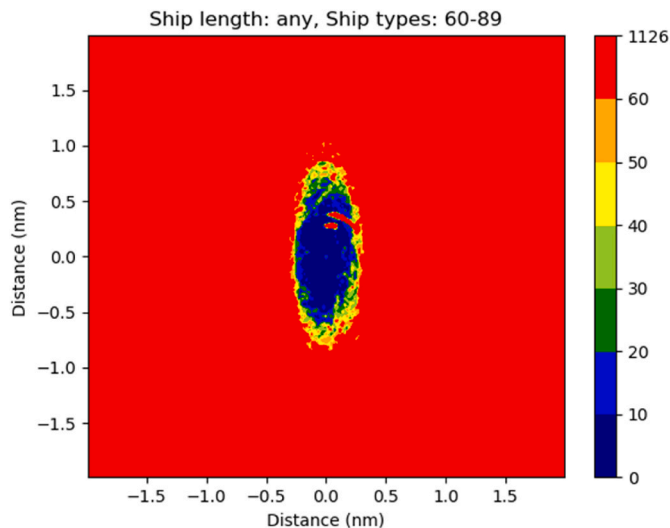


Fig. 3. Relative vessel density contour plot for reference ships with ship types 60-89.

close encounters between ships. The proposed calibrated ship domain is compared with domains generated from earlier studies and is considered superior by its authors.

Zhang and Meng (2019) proposed the concept of probabilistic ship domain, developing an AIS data-driven approach to determine it. These authors state that the boundary between the risk and safe areas is vague and uncertain because of the different navigational behaviours of captains. While the vagueness of the boundary of the probabilistic ship domain has similarities to the boundary depiction in fuzzy domains, the rationale to characterize the uncertainty is different, since in the case of

the probabilistic domain it stems from the frequency of target ships at different distances determined from historical ship trajectories, and in the case of the fuzzy domain, it is a result of the membership functions, which may not be the same when different experts are consulted. Zhang and Meng (2019) elaborated on how the proposed probabilistic ship domain can be used for collision risk assessment, presenting a few case studies from the Singapore Strait.

While numerous models and methods to determine the ship domain exist, and have successfully been applied for different purposes, as previously described in this Chapter, methods to extract empirical ship domains, particularly from AIS data, and the subsequently fitting of analytical ship domain models to those empirical domains, are not commonly found in literature. The small number of parameters, obtained by fitting procedures, needed to describe the analytical models used in this paper contrast with the much larger amount of information required to describe the polygonal empirical ship domains obtained from AIS data. These well-fitted analytical models provide a faster and more computationally efficient way to use violations of ship domain as a collision risk indicator, as demonstrated in Chapter 4.

3. Fitting QSD parameters to AIS data

In this Chapter, a method to fit the parameters of a QSD to an empirical polygonal ship domain obtained from AIS data is proposed. To do so, it is necessary to process AIS data to obtain a record of the positions of nearby ships relative to each reference vessel. In this application, only vessels with Speed Over Ground (SOG) over 2 knots are selected as reference vessels, to potentially exclude vessels that are adrift, anchored or not under command (Wu et al., 2016). Although AIS transmits information on the navigational status of a vessel, this requires the operator to correctly update any change in status manually, which is a naturally error-prone process (Harati-Mokhtari et al., 2007). When monitoring AIS traffic, it is possible to observe vessels that are underway with their AIS navigational status set to *at anchor* or to *moored*, and the opposite is also true.

Within the scope of this paper, reference vessels or ships are those which are taken as a reference to calculate the relative position of all other vessels, in order to obtain the empirical ship domain.

3.1. Extracting relative vessel/traffic density matrices

To generate a record of the positions of surrounding ships relative to a reference vessel, an $n \times n$ square grid is defined for each reference ship, where the length of each side of the grid is denoted as d_{grid} . AIS position reports are processed in the order they were received, and positions of surrounding ships relative to a reference ship are matched to a grid cell. An $n \times n$ matrix is used to keep track of how many times a ship was detected in each grid cell. The pseudo-code of the procedure implemented to perform this task is shown in Algorithm 1. The grid is aligned with the reference ship's heading, except when this information is not available or invalid, in which case the COG is used as the best available approximation. With a slight modification shown in the pseudo-code, the algorithm can be used to obtain both the relative traffic density and the relative vessel density matrices. Extracting the relative traffic and vessel density matrices is the most computationally expensive process of all described in this paper. This step takes approximately 9 h for the entire dataset (413 days), using an Intel i7 processor. Given that the entire dataset covers a period of 413 days, it is possible for the same vessel to enter and exit the geographical area of study several times. To improve the efficiency of the algorithm, two databases are used: one for active ships, which have transmitted AIS messages within a maximum inactive time threshold t_{max} ; another for inactive ships, which have not transmitted any AIS message within the last t_{max} seconds. Because the algorithm loops through the vessels in the active ships database, this strategy improves its computational efficiency. At this point, it is

important to distinguish traffic density and vessel density within the scope of this paper. While vessel density represents the number of reports from vessels within a grid cell over a period of time, traffic density relates to the number of times ships have entered a grid cell over a given time period. In other words, if during a given time period a grid cell was sampled for ten times and the same ship was found to be in the cell all of those ten times, then the traffic density of the cell for that time period would be one, while the vessel density would be ten.

Algorithm 1. Determining relative vessel/traffic density matrices

```

Inputs:
Inputs: File F with decoded AIS sentences
Create database A to store active ships' position, heading, COG, SOG, time since last position
report  $t_A$  and relative position matrix  $M$ , key=MMSI;
Create database I to store inactive ships' relative position matrix  $N$ , key=MMSI;
Create list R to store MMSIs of ships to be removed from A and placed in I;
Create database D to store ship dimensions and type, key=MMSI;
Define maximum inactive time threshold  $t_{max}$ ;
Define length of each grid side  $d_{grid}$ ;
Define number of matrix lines and columns  $n_{grid}$ ;
Define matrix update interval  $t_{updt}$ ;
 $t_0 \leftarrow$  timestamp of first message in F ( $f_0$ );
For each AIS message  $f_i$  in F:
     $t_i \leftarrow$  timestamp of  $f_i$ ;
     $\Delta t \leftarrow (t_i - t_0)$ ;
    If  $\Delta t \neq 0$ :
        For each entry S in database A:
            Update the  $t_A$  field;
            If  $t_A > t_{max}$ :
                Append MMSI of S to list R;
        For each  $r_j$  in R:
            Add entry with MMSI= $r_j$  to database I;
            Delete entry with MMSI= $r_j$  from database A;
        If time since last matrix update  $\geq t_{updt}$ :
            For each entry S in database A:
                 $p_S \leftarrow$  position of S;
                For each other entry T in database A:
                     $p_T \leftarrow$  position of T;
                    If  $t_A^S \neq t_A^T$ :
                         $p_T \leftarrow$  Estimated position of T at the time of  $p_S$ ;
                         $lin_{ST}, col_{ST} \leftarrow$  Matrix coordinates of  $p_T$  relative to  $p_S, f(d_{grid}, n_{grid})$ ;
                        If  $lin_{ST}, col_{ST} \neq$  previous  $lin_{ST}, col_{ST}$  (for rel. traffic density):
                            If  $lin_{ST} \leq n_{grid}$  and  $col_{ST} \leq n_{grid}$ :
                                 $M_S[lin_{ST}, col_{ST}] += 1$ 
                 $t_0 \leftarrow t_i$ ;
            If  $f_i$  message type is 5:
                Insert or update database D values for message MMSI;
            If  $f_i$  message type is 1, 2 or 3:
                Correct ship's position to half-length and half-breadth reference point;
                Insert or update database A values for message MMSI, preserving the matrix in database I if
                an entry exists for that MMSI;
Return list containing matrices  $M$  in database A and matrices  $N$  in database I;

```

All the matrices returned by Algorithm 1 can be added to obtain an overall image of the positions of nearby ships relative to all reference vessels. It is also possible to add matrices obtained from reference vessels within specific length limits or belonging to a given ship type category. To obtain a polygonal ship domain from the added matrices, Python's Matplotlib library is used, which includes a function that returns points of contour lines for specific heights. In this context, heights correspond to the number of times a ship was detected in a grid cell, which is recorded as the values in reference ships' matrices.

Algorithm 1 is applied to data received from ships within a geographical area bound by meridians 9.47° W and 10.5° W, and by

parallels 38.27° N and 39.83° N, recorded by the Portuguese coastal VTS centre (CCTMC) between 05 and 05–2012 and 21-06-2013. Only message types 1, 2, 3 and 5 were used. Message types 1,2 and 3 correspond to Class A AIS position reports, broadcasted at short intervals (between 2 and 10 s), that include the ship's position, course, heading and rate of turn. Message type 5 corresponds to static and voyage related data, broadcasted every 6 min, that includes the ship's name, callsign, IMO number, destination, estimated time of arrival, among other information. Fig. 2 shows a vessel density map obtained using these data. Fig. 3 is a filled contour plot obtained using Matplotlib, that shows the vessel

density around reference vessels with AIS ship types between 60 and 89 (passenger ships, cargo ships and tankers, which originate 90% of the Class A position reports in the dataset, and are found to be less subject to errors and inconsistencies). Distances are expressed in nautical miles (nm). Matrices from 2227 reference vessels are added to produce this plot. Fig. 4 is a contour plot of the traffic density around reference vessels, for the same ship types, using matrices from the same 2227 reference ships. In both cases the t_{updt} parameter is set to 5 s, n_{grid} is set to 201 and d_{grid} is set to four nautical miles.

By slightly modifying Algorithm 1 to increment matrix values only for the positions at which the distance between target ship and reference

ship was minimum, it is possible to obtain a contour plot of CPAs. Such a plot is shown in Fig. 5, for the same ship types, t_{updt} , n_{grid} and d_{grid} as the previous two figures, using matrices from the same 2227 reference ships.

The contours in Fig. 3 take into account the period during which target vessels remain in a certain position, relative to the reference vessel, while the contours in Fig. 4 only reflect the number of target vessels that occupied that position. Low-density areas in the centre of the image, shown in colder colours, are consistent with the notion of a ship domain. Given that the bearing of the CPA between two vessels relative to a reference vessel is perpendicular to the direction of relative motion (assuming rectilinear motion), the high-density areas in Fig. 5 indicate that for most encounters the target ship's course is nearly parallel to the reference ship course, which is consistent with the high number of collision candidates classified as overtaking encounters in the previous chapter.

Matplotlib's contour function returns coordinates of points that define one or more contours that are on the same user-defined level or height. In this context, the function is used to obtain the coordinates of vertices of a polygonal domain, relative to reference ships' positions and headings, where the function's level parameter corresponds to the number of domain violations. These empirical polygonal domains can be based either on the traffic density or the vessel density matrices, but since the definition of ship domain is related to the occurrence of violations of a buffer zone and not necessarily with the amount of time these violations last, throughout this paper traffic density matrices are used for defining empirical ship domains.

3.2. Determining the parameters of the best-fitting QSD

Fig. 6 shows relative traffic density filled contour plots for cargo ships (type 70–79) and tankers (type 80–89), for different ship lengths, with the t_{updt} parameter set to 5 s, n_{grid} set to 201 and d_{grid} set to two nautical miles. The cargo ship and tanker ship types are chosen as research subjects for the remainder of this paper due to the fact that 66.9% and 22.0% of the available AIS Class A position reports are originated from these ship types, respectively. The high-density area in the 200–300-m tankers plot, less than five cables (one cable is equal to one-tenth of a nautical mile) off the bow, is caused by a tug, due to one of the reference ships being towed. Fig. 7 shows the contours for the same vessel types and lengths corresponding to level = 1, which means that the contour line connects cells where only one target ship was detected. Multiple closed contours can be returned by the contour function, as can be observed in this figure. However, not all of them are equally relevant to obtain an empirical ship domain, as not all of them contain the position of the reference ship, with coordinates (0, 0). The following procedure is used to select the most relevant contour:

- All contours which do not contain position (0, 0), where the reference ship is located, are eliminated.
- From the remaining, the one with the largest area is selected. The polygonal contour areas are calculated using equation (4), derived from Green's Theorem, where A^p is the signed area of polygon p , which is defined by n points with coordinates (x_i^p, y_i^p) , with $(x_0^p, y_0^p) = (x_n^p, y_n^p)$. Since the area is signed, depending on whether the points are on a clockwise or counterclockwise order, the absolute value of A^p is used to choose the polygon with largest area.

$$A^p = \sum_{i=0}^{n-1} \frac{(x_{i+1}^p + x_i^p)(y_{i+1}^p - y_i^p)}{2} \quad (4)$$

The results of the application of this procedure to the contours in Fig. 7 are shown in Fig. 8. It should be noted that the small closed

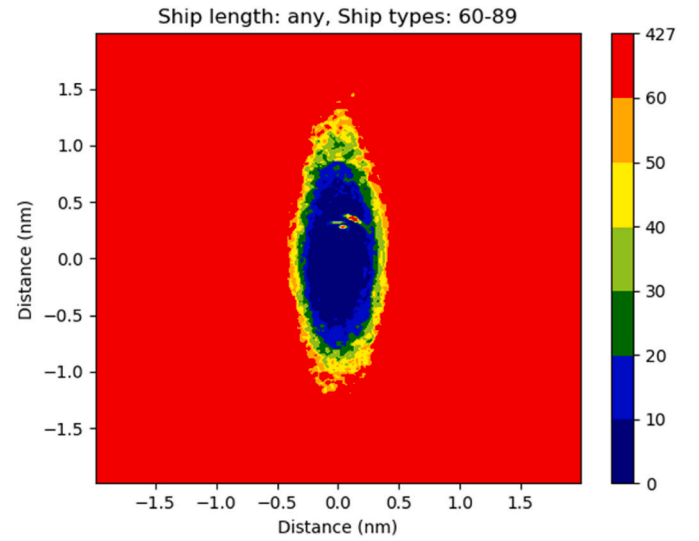


Fig. 4. Relative traffic density contour plot for reference ships with ship types 60-89.

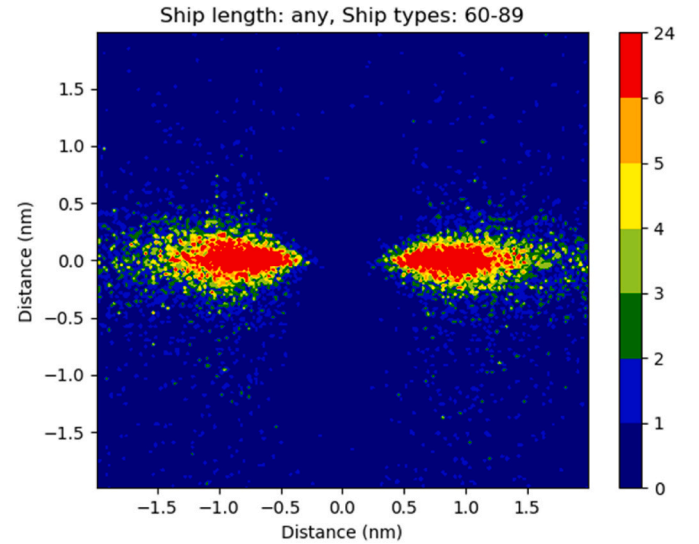


Fig. 5. CPA contour plot for reference ships with ship types 60-89.

contour that corresponds to the high-density area in the 200–300 m tankers plot, caused by a tug towing one of the reference ships, does not influence the chosen contour since it does not contain the point (0, 0).

Finding the parameters of the QSD that best fits the polygonal empirical domain is done by minimizing an error function E^p , given by equation (5), which corresponds to the sum of the squared distances between the polygon vertices (x_i^p, y_i^p) and the corresponding QSD reference points (x_i^q, y_i^q) . Each QSD reference point is collinear with the origin and the corresponding vertex and is located in the same quadrant as this vertex. The procedure used to calculate the QSD reference points is presented in Algorithm 2.

$$E^p(k, R_{fore}, R_{aft}, R_{sb}, R_{port}) = \sum_{i=0}^{n-1} (x_i^q - x_i^p)^2 + (y_i^q - y_i^p)^2 \quad (5)$$

Algorithm 2. Calculating QSD reference points

Inputs:

Polygon vertex (x_i^p, y_i^p)

```

If  $x_i^p = 0$ :
   $x_i^q \leftarrow 0$ ;
  If  $y_i^p > 0$ :
     $y_i^q \leftarrow R_{starb}$ ;
  Else If  $y_i^p < 0$ :
     $y_i^q \leftarrow -R_{port}$ ;
  Else:
    Raise error, contour crosses origin;
Else If  $y_i^p = 0$ :
   $y_i^q \leftarrow 0$ ;
  If  $x_i^p > 0$ :
     $x_i^q \leftarrow R_{fore}$ ;
  Else:
     $x_i^q \leftarrow -R_{aft}$ ;
Else:
   $t \leftarrow \text{abs}(y_i^p / x_i^p)$ ;
  If  $x_i^p > 0$ :
    If  $y_i^p > 0$ :
       $x_i^q \leftarrow (1/(1/R_{fore}^k + (t/R_{starb})^k))^{(1/k)}$ ;
    Else:
       $x_i^q \leftarrow (1/(1/R_{fore}^k + (t/R_{port})^k))^{(1/k)}$ ;
  Else:
    If  $y_i^p > 0$ :
       $x_i^q \leftarrow (1/(1/R_{aft}^k + (t/R_{starb})^k))^{(1/k)}$ ;
    Else:
       $x_i^q \leftarrow (1/(1/R_{aft}^k + (t/R_{port})^k))^{(1/k)}$ ;
   $y_i^q \leftarrow t * x_i^q$ ;
  If  $x_i^p < 0$ :
     $x_i^q \leftarrow -x_i^q$ ;
  If  $y_i^p < 0$ :
     $y_i^q \leftarrow -y_i^q$ ;
Return point  $(x_i^q, y_i^q)$ ;

```

Global minimization algorithms, namely the dual annealing and the differential evolution algorithms, implemented in Python's SciPy library, are used to calculate the QSD parameters that minimize E^p . Both global minimization algorithms provide the same solution, shown in Table 1. Fig. 9 shows the QSDs obtained after parameter fitting, along with the polygonal contours.

3.3. Determining the parameters of the best-fitting modified QSD

A slight modification to the QSD, allowing different values of parameter k for each quadrant, increases the model's flexibility and capacity to adapt to an empirical polygonal ship domain. In this modified QSD model, parameter k in equation (1) becomes a function of x and y , as shown in equation (6). The modified error function is shown in equation (7), with eight parameters that require optimization, and the procedure to calculate the modified QSD reference points is presented in

Algorithm 3.

$$k(x, y) = \begin{cases} k_1, & \text{if } x > 0 \text{ and } y \geq 0 \\ k_2, & \text{if } x \leq 0 \text{ and } y > 0 \\ k_3, & \text{if } x < 0 \text{ and } y \leq 0 \\ k_4, & \text{if } x \geq 0 \text{ and } y < 0 \end{cases} \quad (6)$$

$$E^p(k_1, k_2, k_3, k_4, R_{fore}, R_{aft}, R_{sb}, R_{port}) = \sum_{i=0}^{n-1} (x_i^q - x_i^p)^2 + (y_i^q - y_i^p)^2 \quad (7)$$

The results of fitting the modified QSD to the empirical polygonal domain are again identical between the two algorithms and are shown in Table 2. Fig. 10 shows the QSDs obtained after parameter fitting and the corresponding empirical polygonal contours. The value of the modified error function obtained after fitting is smaller than the value of the error function in equation (5), which indicates a better fitting.

Algorithm 3. Calculating modified QSD reference points

Inputs:

Polygon vertex (x_i^p, y_i^p)

```

If  $x_i^p = 0$ :
   $x_i^q \leftarrow 0$ ;
  If  $y_i^p > 0$ :
     $y_i^q \leftarrow R_{starb}$ ;
  Else If  $y_i^p < 0$ :
     $y_i^q \leftarrow -R_{port}$ ;
  Else:
    Raise error, contour crosses origin;
Else If  $y_i^p = 0$ :
   $y_i^q \leftarrow 0$ ;
  If  $x_i^p > 0$ :
     $x_i^q \leftarrow R_{fore}$ ;
  Else:
     $x_i^q \leftarrow -R_{aft}$ ;
Else:
   $t \leftarrow \text{abs}(y_i^p/x_i^p)$ ;
  If  $x_i^p > 0$ :
    If  $y_i^p > 0$ :
       $x_i^q \leftarrow (1/(1/R_{fore}^{k_1} + (t/R_{starb})^{k_1}))^{(1/k_1)}$ ;
    Else:
       $x_i^q \leftarrow (1/(1/R_{fore}^{k_4} + (t/R_{port})^{k_4}))^{(1/k_4)}$ ;
  Else:
    If  $y_i^p > 0$ :
       $x_i^q \leftarrow (1/(1/R_{aft}^{k_2} + (t/R_{starb})^{k_2}))^{(1/k_2)}$ ;
    Else:
       $x_i^q \leftarrow (1/(1/R_{aft}^{k_3} + (t/R_{port})^{k_3}))^{(1/k_3)}$ ;
   $y_i^q \leftarrow t * x_i^q$ ;
  If  $x_i^p < 0$ :
     $x_i^q \leftarrow -x_i^q$ ;
  If  $y_i^p < 0$ :
     $y_i^q \leftarrow -y_i^q$ ;
Return point  $(x_i^q, y_i^q)$ ;

```

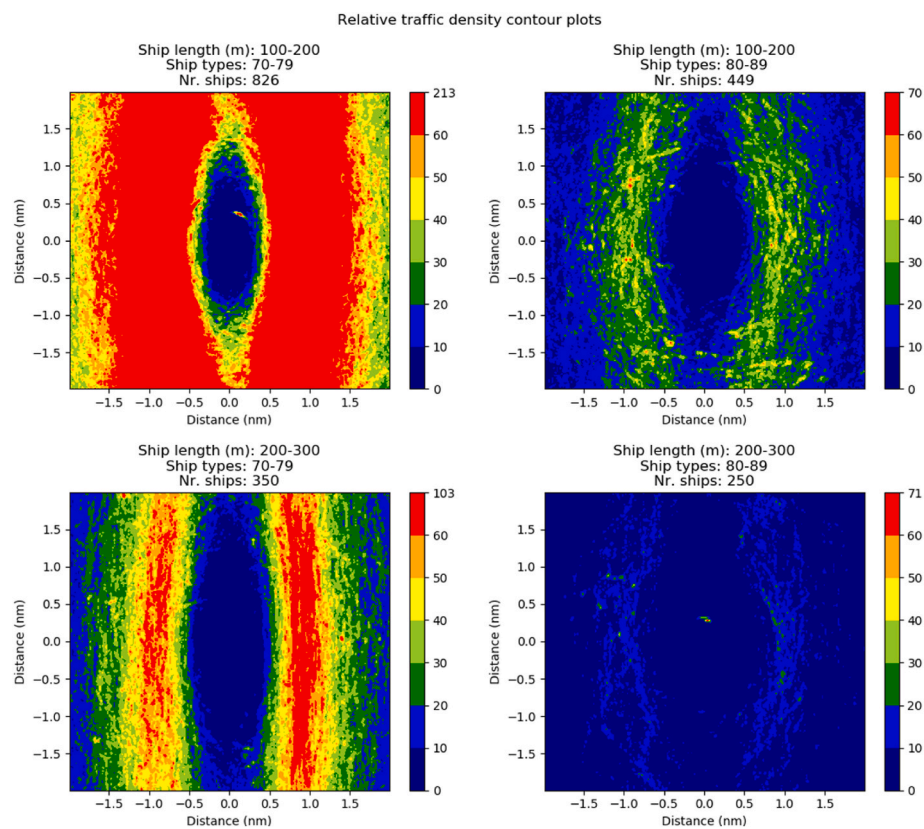


Fig. 6. Relative traffic density contour plots for cargo ships (type 70–79) and tankers (type 80–89), for lengths 100–200 m and 200–300 m.

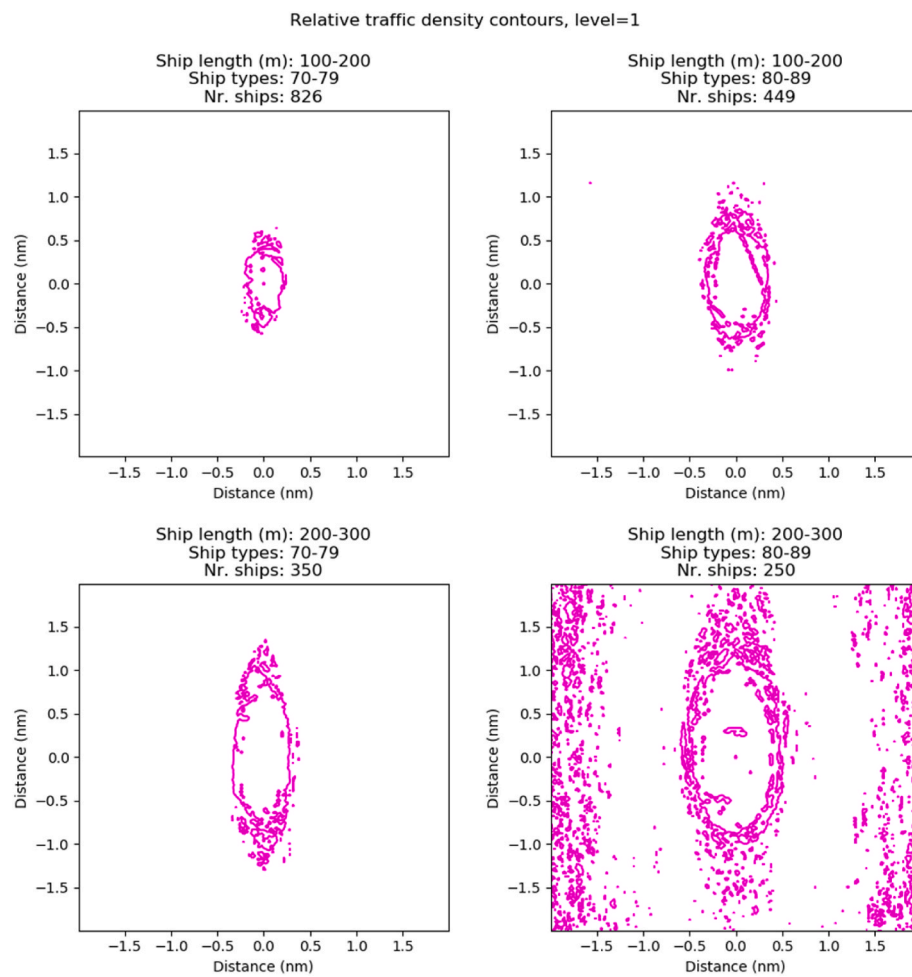


Fig. 7. Relative traffic density contour plots for cargo ships (type 70–79) and tankers (type 80–89), for lengths 100–200 m and 200–300 m, contour level = 1.

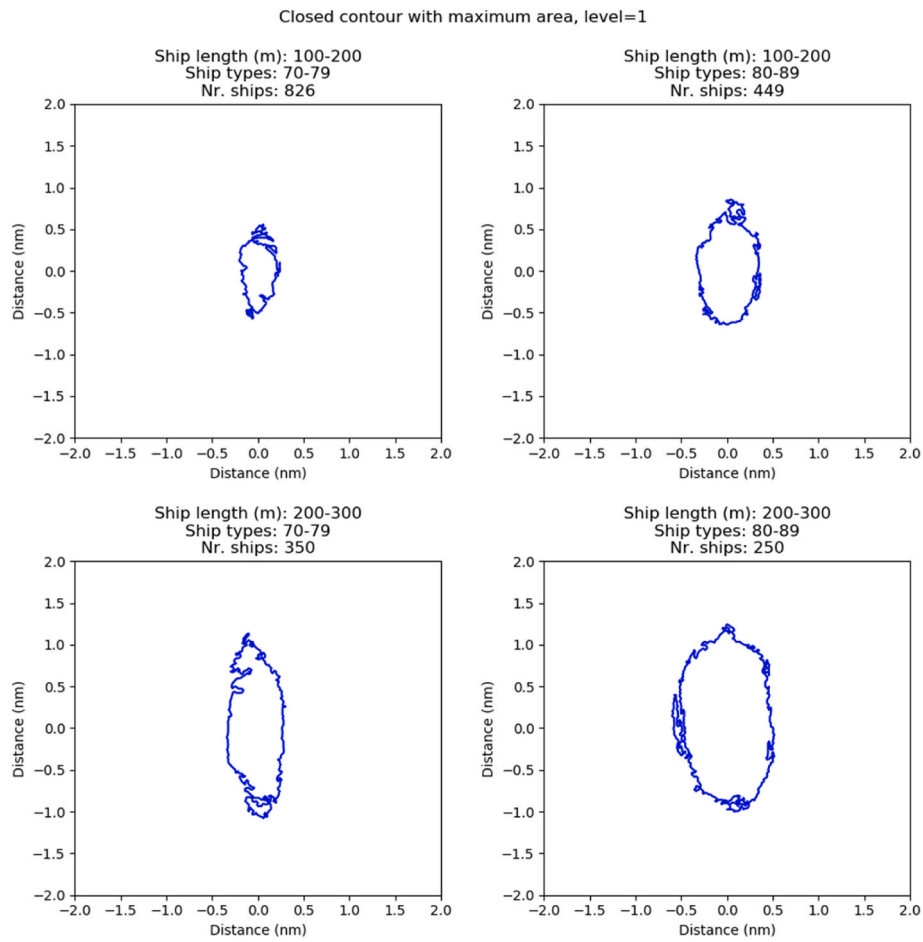


Fig. 8. Closed contours with maximum area for cargo ships (type 70–79) and tankers (type 80–89), for lengths 100–200 m and 200–300 m, contour level = 1.

Table 1

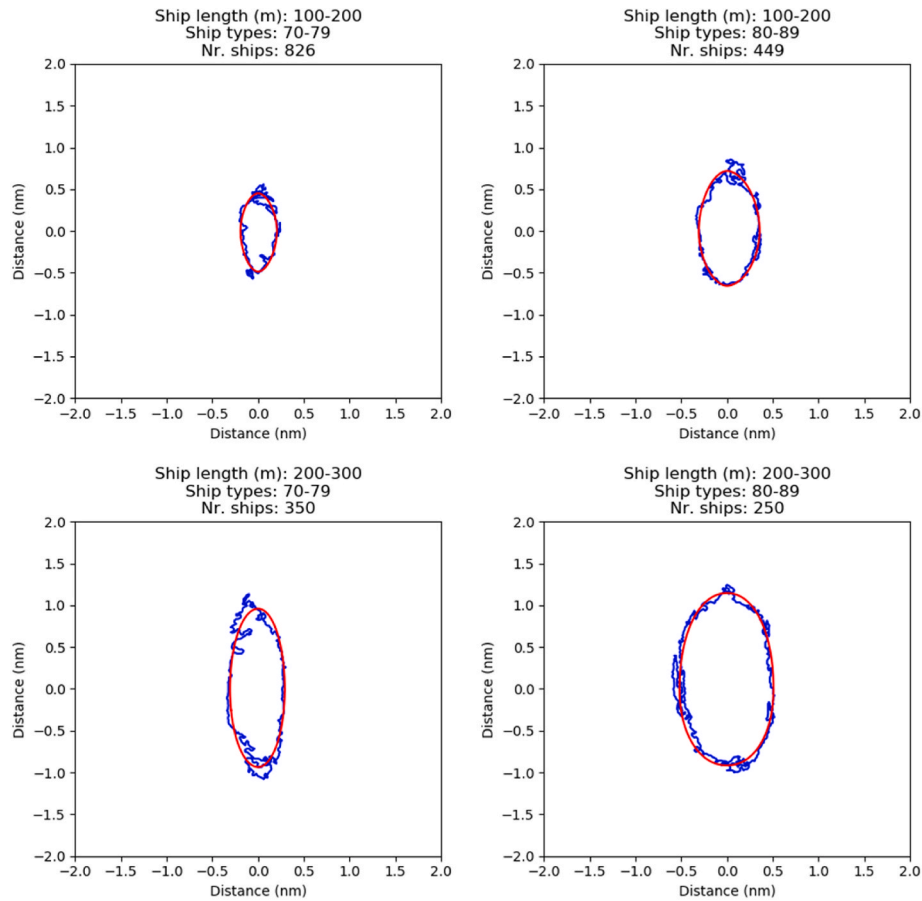
Fitted QSD parameters.

	Length:100–200 m Cargo ships (Type: 70–79)	Length: 100–200 m Tankers (Type: 80–89)	Length: 200–300 m Cargo ships (Type: 70–79)	Length:200–300 m Tankers (Type: 80–89)
Differential Evolution/Dual Annealing algorithms				
k	1.789	1.841	2.007	2.118
R_{fore}	0.451	0.719	0.959	1.146
R_{aft}	0.488	0.657	0.936	0.916
R_{sb}	0.206	0.357	0.291	0.508
R_{port}	0.193	0.309	0.307	0.519

Table 2

Fitted modified QSD parameters.

	Length: 100–200m Cargo Ships (Type: 70–79)	Length: 100–200m Tankers (Type: 80–89)	Length: 200–300m Cargo Ships (Type: 70–79)	Length: 200–300m Tankers (Type: 80–89)
Differential Evolution/Dual Annealing algorithms				
k_1	1.676	2.047	1.614	2.435
k_2	1.011	1.848	2.558	2.791
k_3	2.246	1.965	1.424	1.574
k_4	2.174	1.426	2.148	2.203
R_{fore}	0.446	0.720	0.961	1.129
R_{aft}	0.527	0.645	0.957	0.920
R_{sb}	0.230	0.351	0.282	0.485
R_{port}	0.170	0.322	0.317	0.536

**Fig. 9.** Fitted QSD

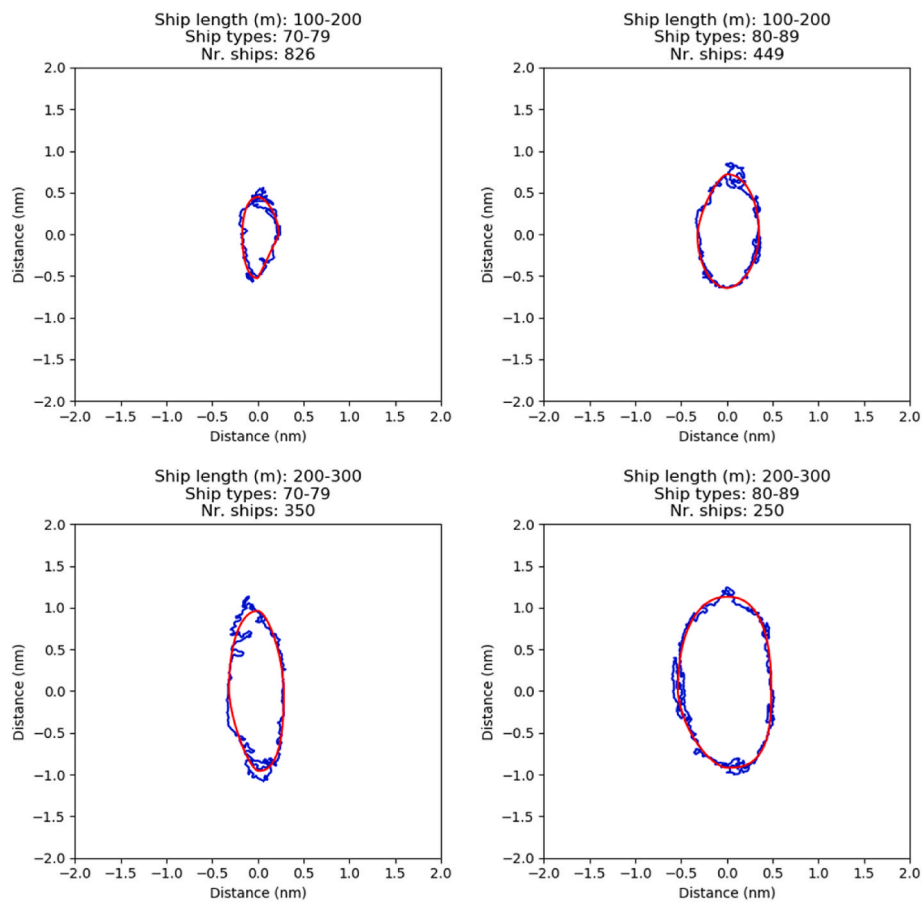


Fig. 10. Fitted modified QSD.

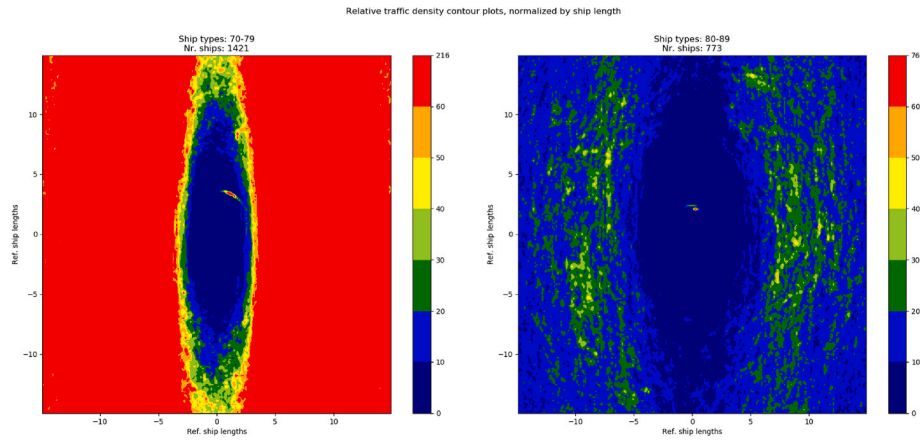


Fig. 11. Relative traffic density contour plots, normalized by ship length, for cargo ships (type 70–79) and tankers (type 80–89).

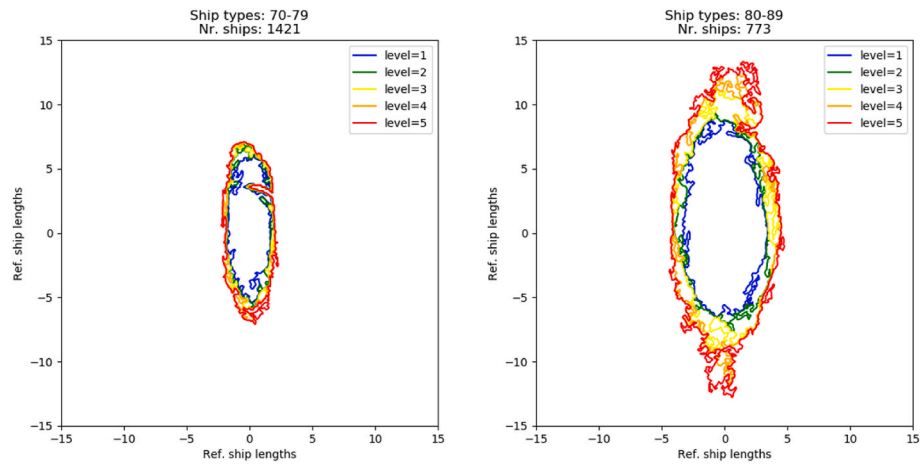


Fig. 12. Polygonal empirical ship domains for cargo ships (type 70–79) and tankers (type 80–89), normalized by ship length, for different contour levels.

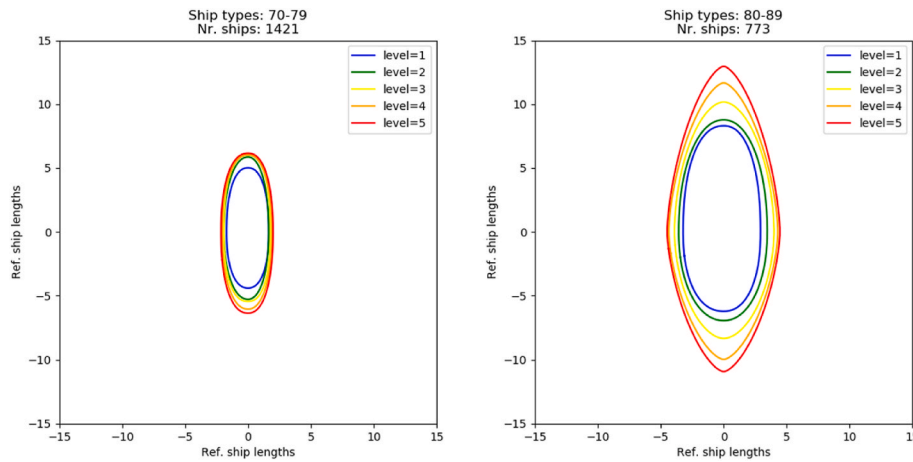


Fig. 13. Fitted QSD for cargo ships (type 70–79) and tankers (type 80–89), normalized by ship length, for different contour levels.

Table 3

Fitted QSD parameters for cargo ships and tankers, normalized by ship length, for different contour levels.

	Level = 1	Level = 2	Level = 3	Level = 4	Level = 5
Differential Evolution algorithm, cargo ships (type 70–79)					
k	2.276	2.127	2.358	2.197	2.275
R_{fore}	5.015	5.865	6.033	6.033	6.162
R_{aft}	4.399	5.282	5.444	6.048	6.358
R_{sb}	1.649	1.653	1.780	1.902	1.988
R_{port}	1.708	1.914	1.921	2.061	2.148
Differential Evolution algorithm, tanker (type 80–89)					
k	2.244	2.089	1.826	1.585	1.551
R_{fore}	8.308	8.774	10.175	11.676	12.973
R_{aft}	6.214	6.943	8.342	9.984	10.946
R_{sb}	2.925	3.460	3.997	4.305	4.469
R_{port}	3.224	3.578	3.915	4.372	4.502

3.4. Determining the parameters of the best-fitting QSD, normalized by ship length

From the observation of Figs. 9 and 10, it is apparent that the ship domain dimensions increase with the increase in ship length. This is expected, and in fact, ship domains found in the literature are often normalized by ship length. It is also clear that the dimensions of tanker ship domains are larger than those of cargo ships with the same length. The procedure described in Algorithm 1 can be used to obtain ship domain matrices normalized by ship length, by assigning the d_{grid} parameter a different value for each reference ship, depending on its length and a k_{length} constant which defines the extents of the grid.

Equation (8) is used to calculate each reference ship's $d_{grid}^{(i)}$ parameter, where $L^{(i)}$ is the length of reference ship i . Fig. 11 shows relative traffic density contour plots for cargo ships and tankers, normalized by ship length, for $n_{grid} = 201$, $k_{length} = 15$ and $t_{updt} = 5$ s. Fig. 12 shows the corresponding polygonal empirical domains, for levels 1 to 5. Fig. 13 shows the optimized QSD obtained for the same contour levels. A set of constraints are used in the minimization procedure, to ensure that the R_{fore} , R_{aft} , R_{sb} and R_{port} parameters for each level take greater or equal

values than those of the previous level, thus preventing a QSD with a lower level from exceeding a QSD with a higher level. Table 3 presents the parameters of the QSDs shown in Fig. 13, obtained by minimizing the error function in equation (5) using the Differential Evolution global minimization algorithm. With the same constraints, the modified QSD models result in the overlapping of domains with different levels, so it is found not adequate for this particular use. It should be noted that the dimension and shapes of empirical ship domains are linked with particular characteristics of the waterways where those ship domains are being practiced, which can explain eventual differences between the results here obtained and the results obtained when studying ship domains in other geographical areas, and waterways with different features.

$$d_{grid}^{(i)} = 2L^{(i)}k_{length} \quad (8)$$

Fig. 14 shows a plot of the variation of the dimensions of the QSD according to ship speed, as per equations (2) and (3). The fitted QSD parameters obtained for different speed ranges, for ship types 70–79 (cargo ships) and contour level 1 are presented in Table 4 and Fig. 15. The corresponding QSD are shown in Fig. 16. While the estimated QSD parameters shown in Fig. 14 increase with speed, the fitted parameters decrease from the 5–10 knots range to the 10–15 knots range. The estimated R_{sb} parameter is greater than R_{port} , which is consistent with experts' preferences (e.g. Silveira et al., 2019), while the opposite is true in the fitted parameters. However, the difference between R_{sb} and R_{port} in Fig. 15 is small, which does not indicate a clear preference for a larger safety distance on one of the sides. The existence of routing measures (Traffic Separation Schemes) in the area of study, and the resulting traffic flows (observable in Fig. 2), favour the occurrence of overtaking encounters, for which the difference between R_{sb} and R_{port} is less relevant compared to other encounter types, such as crossings. This may explain the differences between the generic approach represented in Fig. 14, applicable to all encounter scenarios, and the fitted values shown in Fig. 15, that reflect the specific traffic conditions on the study area. The R_{fore} and R_{aft} fitted parameters seem to converge with the increase in speed, while the estimated parameters slightly diverge. By observing Fig. 16 it is interesting to notice that the largest ship domains occur for the lowest and highest speed ranges, while the smallest ship domains occur for the intermediate speed ranges. This seems to contradict the common understanding that ship domain size typically increases with

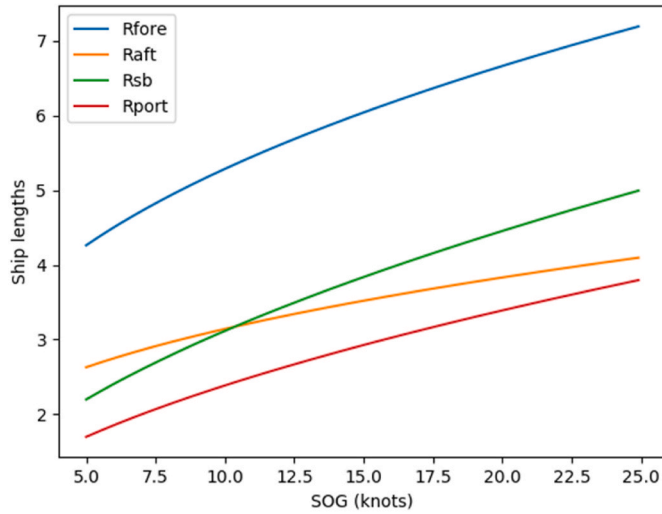


Fig. 14. Estimated QSD parameters, equations (2) and (3).

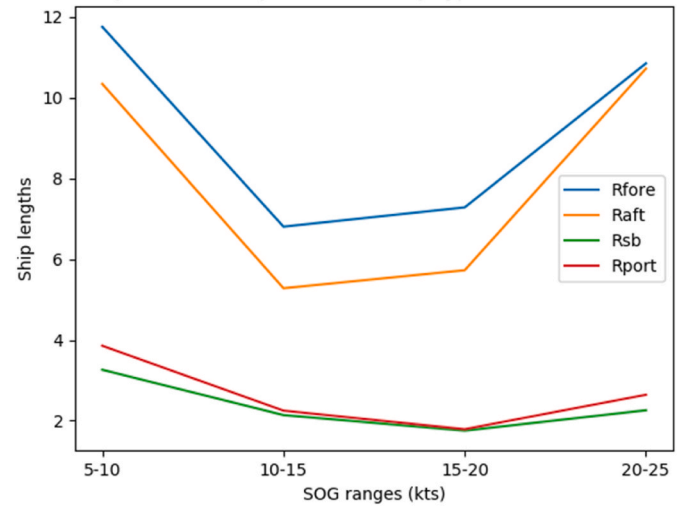


Fig. 15. Fitted QSD parameters for different speed ranges, level = 1, cargo ships (type 70–79).

Table 4
Fitted QSD parameters for different speed ranges, level = 1, cargo ships (type 70–79).

Speed range	k	R_{fore}	R_{aft}	R_{sb}	R_{port}	Nr. ref. ships
5–10 kts	2.321	11.756	10.344	3.262	3.857	541
10–15 kts	1.808	6.808	5.281	2.139	2.249	1138
15–20 kts	2.422	7.285	5.728	1.755	1.790	522
20–25 kts	1.801	10.852	10.722	2.258	2.642	230

speed, as depicted in Fig. 14 and based in the respective equations. In fact, lower speeds, possibly resulting in less manoeuvrability, empirically result in the need to give a wider berth to other vessels.

4. Using violations of QSD as a collision risk indicator

In Chapter 1, deviations from a safety reference level were identified as one of three mechanisms in interpreting risk. In the context of ship collision risk, ship domains, by definition, are such safety reference levels, and therefore are a natural threshold for risk assessment. The different contour levels used to obtain the QSD parameters in Table 3 are related to the number of vessels allowed inside of those contours, and therefore represent different levels of risk perceived by the reference ships.

Fig. 17 shows a heat map of QSD violations, obtained using a Kernel Density Estimation (KDE) algorithm implemented in the QGIS software, inversely weighted by level (violations of level 1 QSD have weight 5, violations of level 5 QSD have weight 1), for reference ships of types 70–79 (cargo ships) with SOG over 5 knots. Fig. 18 shows the same type of plot for ship types 80–89 (tankers). The QSD parameters used to

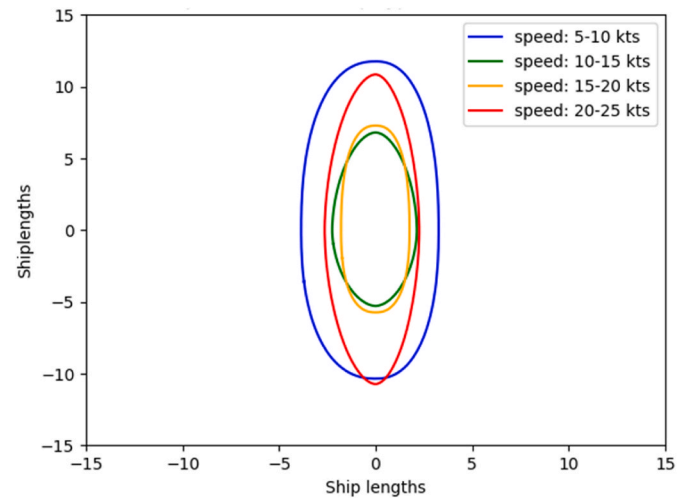


Fig. 16. Fitted QSDs for different speed ranges, cargo ships (type 70–79), level = 1.

determine the violations are the ones presented in Table 3. Regarding Fig. 17, there is a hot spot inside the southbound non-dangerous cargo traffic lane, an area of dense traffic and where ships are bounded by the limits of the traffic lane, and another near the north border of the studied area, which requires further research in order to be explained. Regarding Fig. 18, there is a hot spot on the northbound dangerous cargo traffic lane and another in an area of intersecting routes, observable in Fig. 2, and high fishing activity, identified in the data.

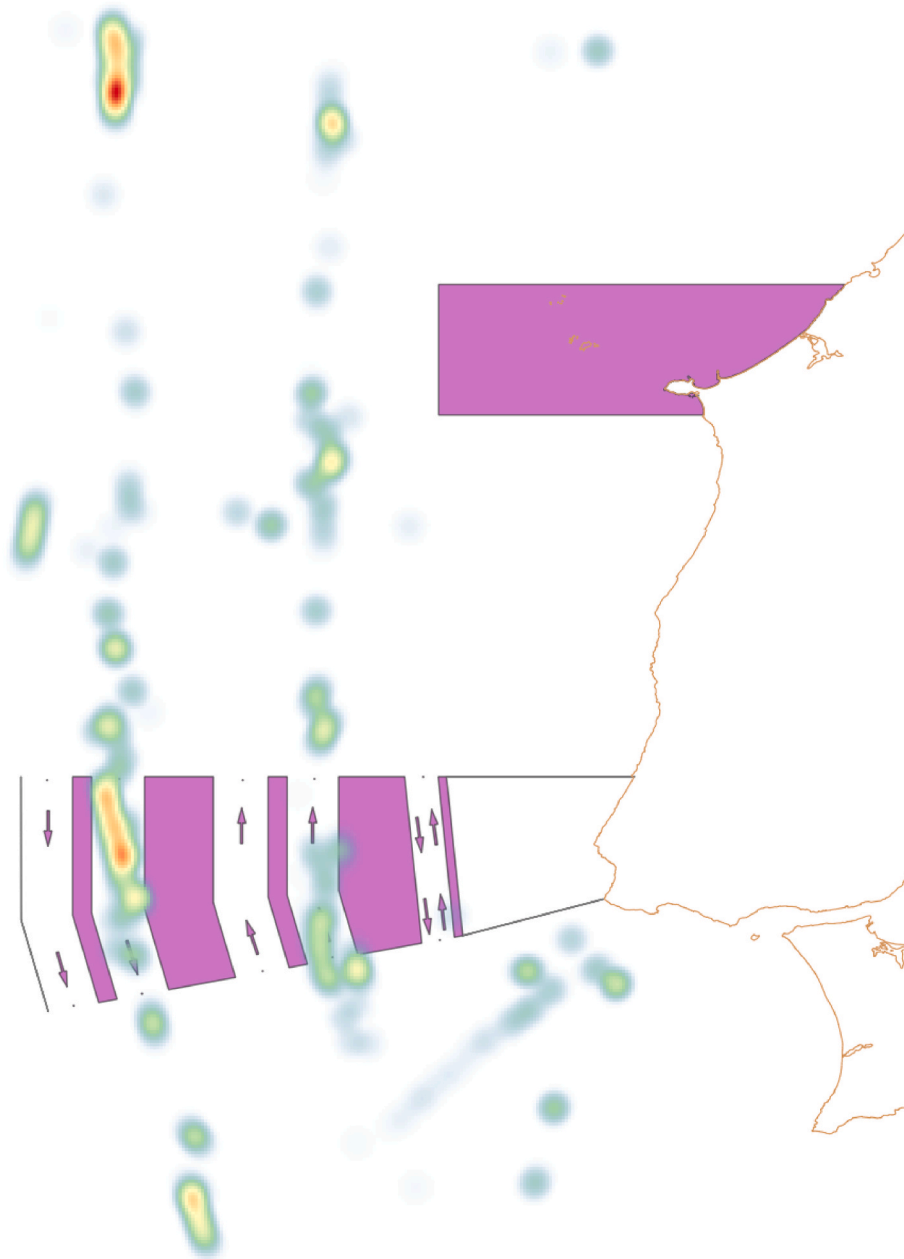


Fig. 17. Heat map of QSD violations, inversely weighted by level, ship type codes 70–79 (cargo ships).

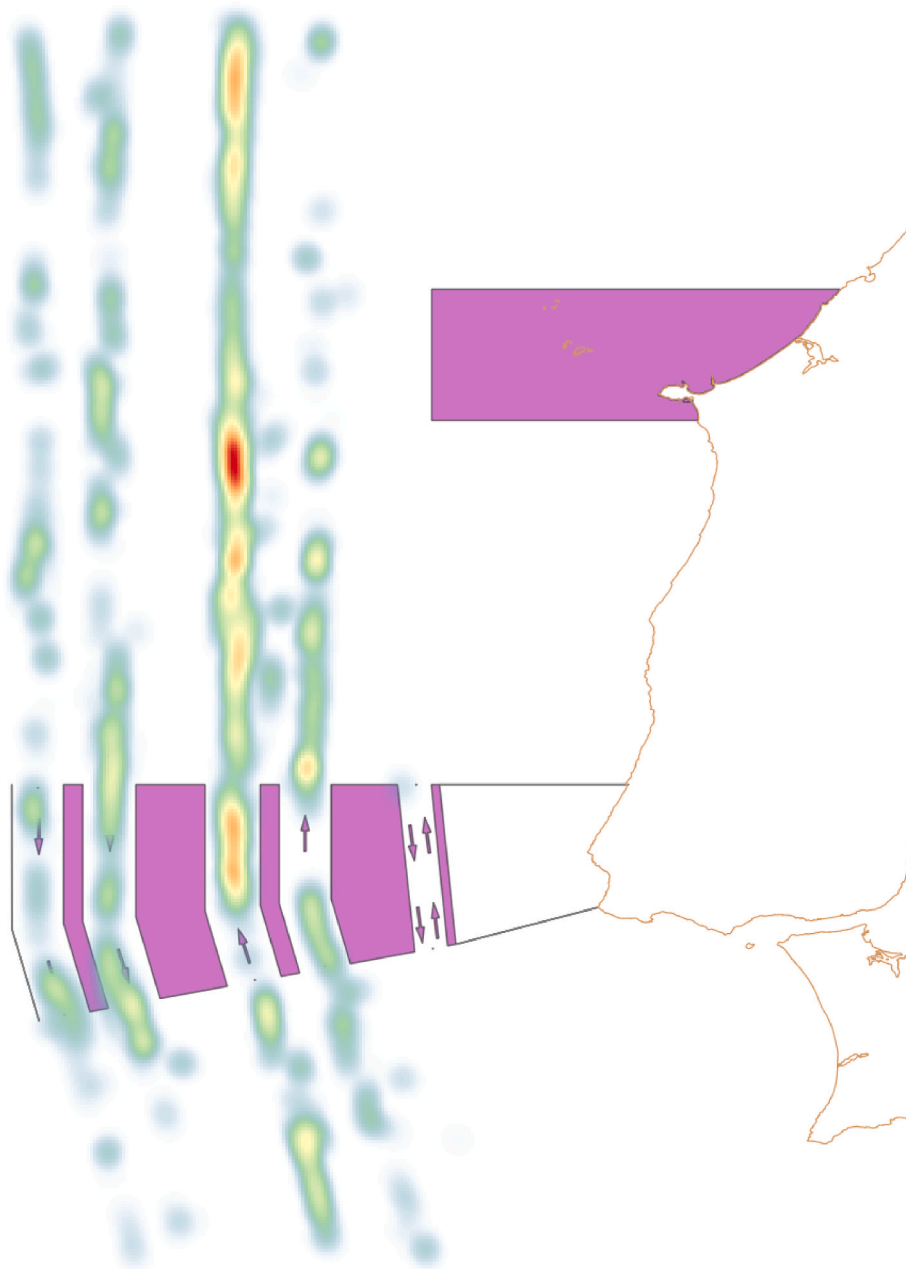


Fig. 18. Heat map of QSD violations, inversely weighted by level, ship type codes 80–89 (tankers).

5. Conclusions

The ship domain concept has been extensively used for studying maritime traffic and ship collision risk. The practice of defining a minimum CPA distance (which corresponds to a circular ship domain) to be observed by officers of the watch is common among shipping companies and masters. However, AIS data-based observations lead to the conclusion that the actual ship domain practised is not circular.

The Quaternion Ship Domain is an analytical model that can be adapted to various shapes and sizes by changing its five parameters. In this paper, a method to fit the QSD model to a polygonal ship domain obtained from AIS data is proposed and successfully applied. Comparing the fitted QSD parameters for cargo ships and tankers with different lengths leads to the conclusion that there is a noticeable increase in the radii as length increases, and also that tankers tend to have a larger ship domain than cargo ships. A modified version of the QSD model is proposed, that allows for different shape parameters for each domain

quadrant. This allows the analytical domain to better fit oddly shaped empirical domains, however, the extra flexibility of this model may cause the limits of the optimized modified QSD domains with different levels to cross each other.

The limits of the QSDs obtained for different levels are linked to the number of target ships allowed inside the respective polygonal empirical domains, and therefore represent a measure of risk as perceived by the officers in the reference ships. Visualization tools are used to identify geographical areas where violation of the QSDs, inversely weighted by their levels, are more significant, and therefore where collision risk is greater.

Although in this paper special relevance is given to the deviation from a safety reference level mechanism when using ship domains to interpret collision risk, close-quarters situations resulting from a domain violation can also be thought of as ambiguous situations, which are another of those mechanisms. This is because the COLREGs state that, if necessary, ships may depart from the rules to avoid a collision, therefore

departing from expected behaviours when attempting last resort collision avoidance measures. However, one can reason that these potential less predictable behaviours at close quarters are already taken into account and reflected in the ship domains that are used in practice.

CRedit authorship contribution statement

P. Silveira: Conceptualization, Methodology, Software, Data curation, Writing – original draft. **A.P. Teixeira:** Conceptualization, Methodology, Writing – review & editing, Supervision. **C. Guedes Soares:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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